Control Stage (CTRL) — Verification Plan

UVM Verification Environment | Version 0.1 (draft)

Project: ctrl_uvm_environment | Source specs: Control Stage, Cache Unit | October 2026

1. Overview

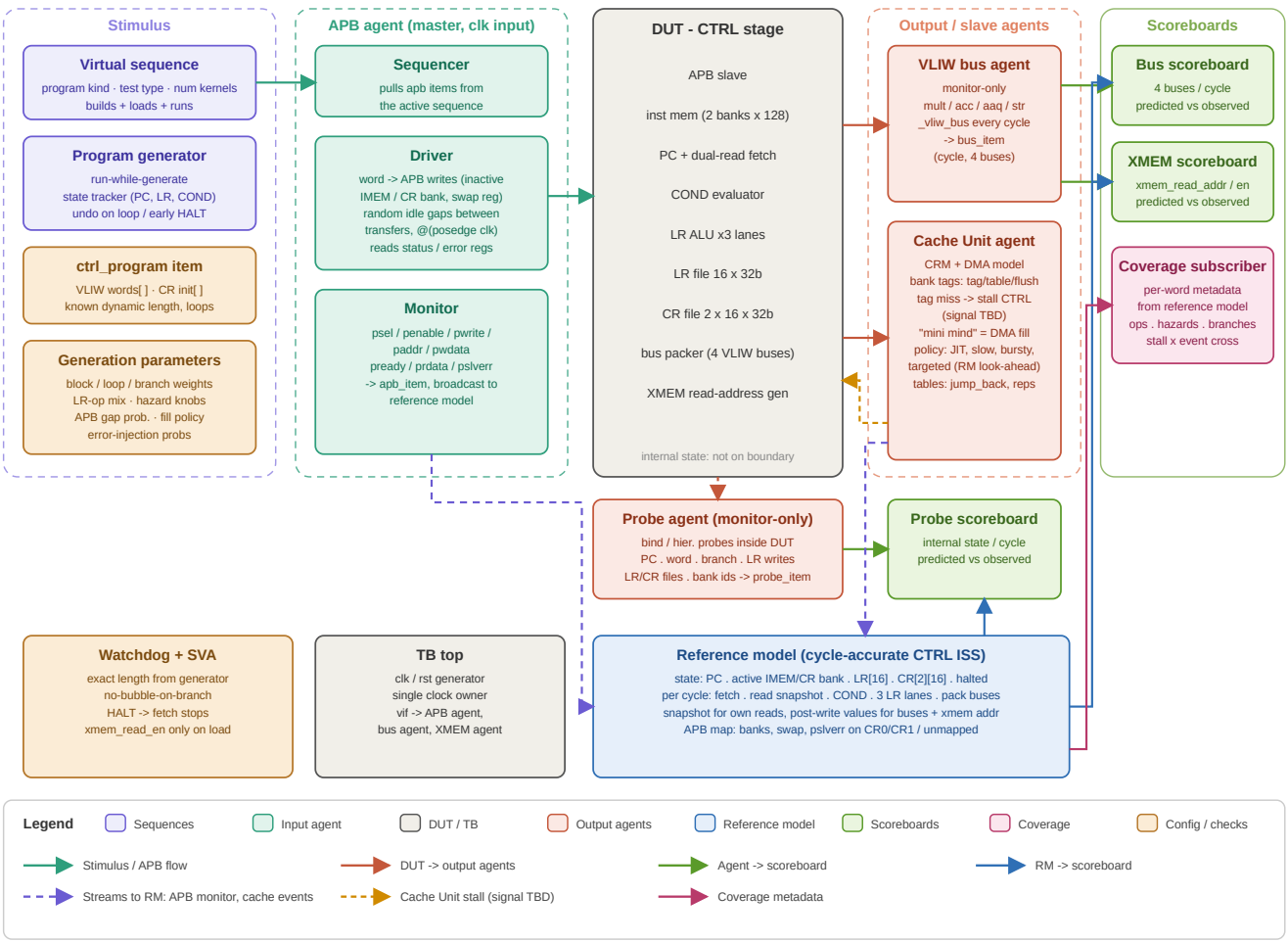
The Control (CTRL) stage is the first stage of the IPU pipeline. It owns the program counter, a double-buffered instruction memory (2 banks × 128 decoded VLIW words), the CR and LR register files, the branch resolver and three parallel LR ALU lanes. It drives four per-stage VLIW buses (`mult/acc/aaq/str_vliw_bus`) and the XMEM read address (`xmem_read_addr`, `xmem_read_en`). A RISC-V host configures it through an **APB slave** port.

This plan covers **CTRL only**. The downstream stages (MULT, ACC, AAQ, STORE) and XMEM data return are not in the environment: their buses and the XMEM address port are *monitored*, not consumed.

1.1 What makes CTRL hard to verify

- **Programs are the stimulus.** The DUT's behaviour is defined by the VLIW program in its IMEM. Random programs that are *legal and known to stop* are the main source of coverage (§4).
- **Two read semantics in one cycle.** CTRL's own reads (LR-ALU sources, branch operands) use the *prior-cycle snapshot*; the values forwarded on the buses and used for `xmem_read_addr` are the *post-LR-write* values.
- **No-bubble branches.** Both `PC+1` and the branch target are fetched in parallel; the correct word must always be present next cycle.
- **Double-buffered configuration.** The host writes the inactive IMEM/CR bank over APB while the active bank executes.
- **Cache Unit stalls.** The Cache Unit (CRM) stalls the controller when the bank holding the requested XMEM address is not ready. Stalls must change timing only, never results (see Q1, §9).

2. Environment Architecture



- The **virtual sequence** generates a random program that is known to stop (§4.4), loads it through the APB agent into the inactive bank, swaps, and lets it run. It is the only place that knows the whole program.
- The **Cache Unit agent** sends stall events, bank-tag changes and table configuration to the reference model, so predicted timing includes the stalls the DUT really saw.
- The **probe agent** samples CTRL's internal signals every cycle; the **probe scoreboard** compares them with the reference model's internal state.
- The **reference model** is a cycle-accurate model of CTRL. It learns the program **only** from the APB monitor (what the DUT actually received), never from the generator, and predicts the four buses and the XMEM address for every cycle.
- Each **scoreboard** receives predicted items from the reference model and observed items from the relevant monitor, and compares them.
- The **coverage subscriber** receives per-word metadata from the reference model and samples the functional covergroups.
- The Cache Unit agent models the stall source from XMEM tag state. The clock and reset are owned by the TB top. The APB agent only *receives* `clk` through its virtual interface.

3. Agents

All agents use one generic `dv_agent` pattern (as in the RX DMA environment): sequencer + driver + monitor, with `is_master` selecting driver/sequencer. Output agents are monitor-only.

3.1 APB agent (master, clock as input)

Sequencer

Pulls `apb_item`s from the active sequence and forwards them to the driver one at a time.

Driver

Drives `apb_psel / penable / pwrite / paddr / pwdata` and honours `pready` wait states. It only waits on `@(posedge vif.clk)`; it never generates the clock. The idle gap between transfers is randomised, three modes selected once per test via config_db:

- Back-to-back — no idle cycles (`gap_probability = 0`)
- Sparse random — idle cycles with probability 75%
- Deterministic — fixed pattern: transfer, idle, idle, idle

Monitor

Collects every APB transfer into an `apb_item` (address, data, direction, `prdata`, `pslverr`) and broadcasts it to the reference model. Always active.

3.2 Output agents (slave, monitor-only)

Agent	Signals monitored	Item produced
VLIW bus agent	`mult_vliw_bus, acc_vliw_bus, aaq_vliw_bus, str_vliw_bus` every cycle	`bus_item`: cycle number + four bus values
Cache Unit agent	`xmem_read_addr, xmem_read_en`; drives the stall and feeds the reference model (see 3.4)	`xmem_item`: address when enable is high, plus stall start / end events

> **Bus format is TBD**
> The spec leaves `MULT_BUS_W / ACC_BUS_W / AAQ_BUS_W / STR_BUS_W` as "impl". The bus agent samples them as raw vectors; a `bus_layout` class (slot fields + resolved CR/LR operand fields) is needed before the scoreboard can compare field-by-field.

3.3 Probe agent (monitor-only, internal signals)

The boundary agents only see the buses, the XMEM port and APB. A wrong PC, a wrong LR write that is never read again, or a wrong branch decision can stay invisible there. The **probe agent** looks **inside** the DUT: it is a passive

monitor attached with `bind` (or hierarchical references) to CTRL's internal signals. It never drives anything.

Probed signals

Group	Signals (spec names)	Used to check
Fetch	`PC, next_pc, imem_read_addr_a, imem_read_addr_b, imem_a, imem_b, imem_rd_sel, inst_vliw`	PC trace; `inst_vliw == imem[active][PC]`; the parallel read of `PC+1` and the target
Branch	`branch_taken`, resolved target	Taken / not taken and target for every COND op
LR write lanes	`lr_write_en[3], lr_write_idx[3], lr_write_data[3]`	Each lane's write, per cycle, before it can reach any bus
Register files	`LR[0..15]`, active `CR[0..15]`	Full architectural state each cycle; `CR0 = 0`, `CR1 = 1`
Control state	active IMEM bank, active CR bank, `fetch_stopped` (after HALT)	Bank-swap timing; fetch stops after HALT and restarts at PC 0
Operand paths	snapshot values read by the lanes / COND, and the post-write values packed on each bus	The two read semantics (snapshot vs. post-write) side by side

What it gives the environment

- **Early, precise failures.** Each cycle it builds a `probe_item` (cycle, PC, next PC, word, branch result, three LR writes, LR file, bank IDs, `fetch_stopped`). The probe scoreboard flags the first internal divergence at the exact cycle and unit, instead of a bus mismatch many cycles later.
- **Visibility of masked bugs.** A wrong LR write that no later word reads, or a wrong `PC` that fetches an identical word, still shows up.
- **Real hazard coverage.** The coverage subscriber samples hazards from what the DUT *actually did* (LR written in cycle N and read in cycle N), not only from what the generator intended.
- **Assertion hooks.** The SVA checks of §6.3 bind to the same probe interface.

Implementation notes

- All RTL hierarchy names live in **one** mapping file (`ctrl_probe_if`). If the RTL is renamed, only that file changes.
- Sampled once per cycle through a clocking block, after the cycle has settled.
- Enabled by `PROBE_EN`; disabled automatically for runs where internals do not exist (e.g. gate-level).
- Internal names come from the note in spec §3 (`PC, next_pc, branch_taken, imem_*, inst_vliw, lr_write_*`); see Q9.

3.4 Cache Unit agent (CRM + DMA model) — the "mini mind"

The Cache Unit sits between XMEM and the IPU. It is the only block that knows **which addresses XMEM currently holds**, so it is the one that stalls the controller. The agent is therefore a **behavioural model of the Cache Unit**, not a random wait generator: a stall appears only when the requested bank is not ready.

State and rules (from the Cache Unit spec)

- Per bank: `{tag (14 bit), table_id, flush}`, `NUM_BANKS = 16`, 1024 rows per bank. `tag = FFFF` means free / not ready.
- Per table: `table_id, type, size, banklist, jump_back, repetitions`. For the controller only read tables matter (`read_only`, `scratch_pad`).
- On every access (CRM rule): `bank_pos = offset[19:10] % NUM_BANKS`, `tag = {TABLE_ID, offset[19:10]}`. If `tag != bank.tag` or `bank.table_id != TABLE_ID` → **stall the controller**; release when the DMA model publishes the matching tag.
- Bank release: when `offset[9:0] >= jump_back` and `offset[19:10] != 0`, the previous bank is freed (`tag = FFFF`); `offset == FFFF` frees all banks.

The "mini mind" = the DMA fill policy

Stall length is not drawn from a dice roll. It *emerges* from **when the DMA model publishes each bank's tag** after the bank becomes free. The mini mind is: **fill mode + short history + seeded RNG**, re-picked per table or per K fills.

Fill mode	Behaviour	What it hunts
`PREFETCH_AHEAD`	DMA always one or more banks ahead; tags are published before they are needed	No-stall baseline
`JUST_IN_TIME`	Fill starts only after the first request for the bank → one stall per bank	Stall / release on every bank boundary
`RANDOM_LATENCY`	Fill latency random in [min, max]	General robustness
`SLOW_FILL`	Fill latency of the order of a full bank (≈1024 rows, hundreds of cycles)	Long-stall tolerance, watchdog margin
`BURSTY`	Several banks published together, then a gap	Stall followed by a long no-stall run
`FIRST_PAGE_LATE`	First bank of a table is late (as in the Layer1 timing example)	Stall right at kernel start
`TARGETED`	Uses the reference model's look-ahead: delays exactly the bank the next load will hit — in a branch word, in a word whose offset LR is written in the same word, in the last word before HALT, at the `jump_back` boundary	The corner cases the program really contains

Everything is seeded, so a failing fill pattern replays exactly. The table configuration (ids, banklists, `jump_back`, `repetitions`) arrives through config_db from the virtual sequence; in the real system the RISC-V sets it up over `riscv_cfg_bus`, which is outside the CTRL environment.

Feeding the reference model

The Cache Unit agent has a **monitor that broadcasts to the reference model** through an analysis port, like the APB monitor does. The model cannot predict stalls from the program alone, because stall timing depends on the Cache Unit's bank state and the DMA fill policy. It receives:

- **Stall events** — stall start and stall end, with the address and bank that caused them.
- **Bank / tag state changes** — tag published, bank freed, free-all — so the model's view of XMEM matches the agent's.
- **Table configuration** — once per kernel.

The model uses these to hold PC, word and buses during a stall. In the other direction, the model gives the agent a look-ahead of upcoming load addresses (used by `TARGETED`).

Two consequences for the program generator

- **Load addresses must be legal for the table.** A load must stay inside the table `size`, walk banks in order, and may revisit a freed bank only on a new repetition. A load to a freed bank that DMA will never refill would hang the DUT forever, so that is a **generator bug**, not a DUT bug. The generator's state tracker knows every load address exactly (`CR + LR`), so it rejects a word whose address would be illegal (§4.4).
- **The watchdog must include fill latency** (§4.4).

Open points on the stall path (Q1)

The stall *source* is now clear (Cache Unit / CRM). Still undefined in the CTRL spec: (a) the CTRL port list has no stall / ready input, so it is unknown how the stall reaches the controller (held `xmem_read_en`, a `stall` wire, …); (b) CTRL only outputs `xmem_read_addr`, while the CRM needs `table_id` plus `array_id + offset` — where does `table_id` come from? Until then the agent drives a placeholder `stall` signal.

4. Sequences and Items

4.1 ctrl_word — one VLIW word

IMEM stores **already-decoded** words, so the item mirrors the slot layout. CTRL only demultiplexes it.

Slot	Fields (randomised)	Constraints
COND (×1)	`op` ∈ {NOP, BEQ, BNE, BLT.S, BGE.S, BR, HALT}; `reg1, reg2` (LR/CR); `label` or `reg` (BR)	Branch target must be a legal word in the loaded program. `BR` uses the low 7 bits of the register. `HALT` only as a terminator.
LR (×3 lanes)	`SET, ADD, SUB, INC.MODP2, INC, DEC, ADDB, ADDBI`; `dest, src_a, src_b / imm / k`	No two lanes write the same destination; `SET` imm is signed 4-bit; `INC.MODP2` k ∈ [1,9]; `ADDB/ADDBI` dest is an even `LRDn` pair.
LOAD	NOP or `{base_idx (CR), offset_idx (LR)}`	Resulting address must fit `XMEM_ADDR_W`.
MULT / ACC / AAQ / STORE	Opaque slot payload + the CR/LR indices CTRL must resolve for that stage	CTRL does not decode the payload, only forwards it, so it is random. Indices are random over all 16 LR / 16 CR so every operand path is covered.

STORE slot
ADR 0007 removed the STORE slot, but the Control Stage page still describes `str_vliw_bus` (open question Q2). The plan keeps it as a parameter, `HAS_STORE_BUS`.

4.2 ctrl_cr_config and ctrl_program

- `ctrl_cr_config` — values for `CR2–CR14` (and `CR15`, the conventional DS half) of the inactive CR bank. `CR0 = 0` and `CR1 = 1` are hard-wired and never written, except by error injection.
- `ctrl_program` — one whole kernel: `words[0..127]`, `cr_config`, and what the generator learned while building it (used for coverage, the watchdog and the Cache Unit table config): exact dynamic word count, loops seen, taken branches, `BR` targets, load addresses.

4.3 APB sequences

Sequence	Description
`apb_load_program_seq`	Writes every word of a `ctrl_program` into the **inactive** IMEM bank, writes the CR config into the inactive CR bank, then writes the bank-swap registers.
`apb_status_seq`	Reads status/bank-ID registers while the kernel runs (must add no wait cycles to the pipeline).
`apb_error_seq`	Error injection: write `CR0` / `CR1`, write a read-only register, access an unmapped address — all expect `pslverr`.
`apb_preload_next_seq`	Loads the *next* kernel into the new inactive bank while the current one runs.

4.4 ctrl_virtual_sequence — the program-driven test controller

This is the heart of the environment. The program is the stimulus, and it has only **two limits: it must be legal and it must stop.** Inside those limits it is fully random: any LR op, any COND op (including `BR` whose target comes from an LR), any label, loops entered from anywhere. There is no AST, no control-flow graph and no loop-guard rule. The reference model is not involved in deciding that a program stops.

Why the generator must follow the values

Whether a branch goes to `PC+1` or to its label depends on LR values, and a `BR` goes wherever an LR points. For an **unrestricted** program, the only way to *know* that it ends is to follow those values while the program is built. (If the generator may not follow values, the program must be restricted instead — forward branches and guarded loops — and then it can no longer contain arbitrary loops or computed jumps.) CTRL's control flow depends only on `PC`, `LR[16]` and the fixed CR values, never on XMEM data, stalls or APB timing, so following those values gives an exact answer.

Execution-guided generation

The generator contains its own small **state tracker** (`ctrl_prog_tracker`: `PC`, `LR[16]`, the 3 LR lanes, the COND evaluator, the snapshot rule). It is a separate class from the reference model and is built only from the ISA text.

Step	What happens

1 — Start	Pick a requested length *T* (executed words) and the CR values. `PC = 0`, LR in its starting state. Word 127 is `HALT`.
2 — New word	When execution reaches a word that does not exist yet, create it **fully at random** but legal (see below), then execute it in the tracker. Existing words are executed as they are.
3 — Finish	When *T* words have executed, the next new word is forced to a jump to word 127 (e.g. `BEQ r r` to `HALT`). A stopping continuation therefore always exists.
4 — Failure	Fail if the state `(PC, LR[0..15])` repeats (provably infinite), if the path exceeds a step limit, if execution reaches `HALT` before *T*, or if a load address is illegal for the table (§3.4).
5 — Undo	On failure, go back to the most recent new word, forget it and everything after it, and re-roll that word. After a few tries on one word, go back to the one before.
6 — Fill	Words that were never reached are dead code. Fill them with random legal words or `HALT`; they cannot change behaviour.

The result is a program whose executed path is known exactly: its dynamic length, its loops, every taken branch and every `BR` target. Words are created only when execution needs them, which is why loops and jumps to earlier code appear naturally: a loop is a set of existing words executed again with new LR values, and the tracker watches it until it exits.

Legality rules (the only restrictions on a word)

- No two LR lanes in one word write the same register, counting `LRD` pair aliasing.
- `ADDB/ADDBI` destination is an even `LRDn`; `SET` immediate is signed 4-bit; `INC.MODP2` $k \in [1, 9]$.
- Labels and `BR` targets are word addresses 0–127 (low 7 bits).
- CR0 and CR1 are never written.
- Load addresses (`CR[base] + LR[offset]`, post-write) stay inside the table the Cache Unit agent knows (§3.4), every time the word runs.

Measured on a prototype

A C model of the ISA, 2000 programs per run, each with a requested length *T* chosen at random:

Requested length T	Programs generated	Avg executed words	With a loop
20 – 500	1999 / 2000 (99.95%)	346	96%
20 – 5,000	1719 / 2000 (86%)	872	98%
500 – 5,000 (more retries)	1295 / 2000 (65%)	1,949	100%

Every finished program was run again from scratch and halted at exactly the predicted step count (0 mismatches in 5,000 programs). A failure means the generator gave up after its work budget; long loop-heavy programs need hundreds of undo steps. For comparison, generating random words and then checking accepted 39% of programs but they averaged 17 executed words.

Test flow

Phase	Action
0 — Configure	Randomise `ctrl_gen_parameters` and the Cache Unit policy (fill modes, latencies, table configs) once; publish via config_db.
1 — Generate	Run execution-guided generation. The result is a program **known to stop**, with its exact dynamic length. Nothing has been sent to the DUT or the reference model yet.
2 — Load	`apb_load_program_seq`: program into the inactive IMEM bank, `CR2`–`CR14` into the inactive CR bank, then swap. The new active bank starts at PC 0 *(start mechanism: Q4)*. The reference model learns the program from the APB monitor.
3 — Run	While the kernel executes: optionally preload the next kernel (`apb_preload_next_seq`), read status, inject APB errors. The Cache Unit agent stalls the controller whenever a requested bank is not ready.
4 — Finish	Wait for `HALT` to retire; the watchdog uses the **exact** dynamic length from phase 1 plus Cache Unit fill latency. A timeout is a DUT bug or a mismatch between the tracker and the DUT. Check scoreboards, sample coverage, drop objection.

`test_type`: **REGULAR** — one program (fast). **REAL** — 10–100 kernels back-to-back, each preloaded into the inactive bank while the previous one runs, then swapped.

`program_mode`: `FREE` (fully random), `LONG` / `SHORT` (requested length *T*), `PINNED` (some words fixed by a directed test, the rest random and still known to stop), `BANK_FULL` (uses all 128 words).

Starting state and the tracker

- The check is exact for one starting LR state. Whether the LRs are cleared at kernel start is not in the spec (Q10). In a back-to-back test the generator must use the LR values left by the previous kernel, or zeros if the DUT clears them.
- The tracker is a second copy of the instruction semantics. If it differs from the DUT, a generated program might not stop in the DUT. Check it once, offline, against the reference model on a few thousand programs. During tests the two are not coupled.

4.5 Generation parameters

Parameter group	Controls
Program kind weights	Relative probability of each `program_kind`
Requested length	Range of *T* (executed words), program mode weights
Generator effort	Retries per word, work budget, step limit per attempt
COND mix	Weights of NOP / BEQ / BNE / BLT.S / BGE.S / BR; label distribution
LR-op mix	Weight per LR opcode; lanes active per word (0–3)
Hazard knobs	Probability of same-word write→bus-operand, write→load-offset, write→branch-operand
Valid / gap probability	APB idle-gap mode (back-to-back / sparse / deterministic)
Cache Unit policy	Fill-mode weights, fill latency ranges, phase length K; table config (banks used, `jump_back`, `repetitions`, size)
Error injection	Probability per kernel of each APB error sequence
Kernel count	Min/max kernels in a REAL test

5. Reference Model

A cycle-accurate model of CTRL (not transaction-level: the spec's behaviour is defined per cycle). It receives APB items from the APB monitor (bank writes and swaps) and **stall events, bank-tag changes and table configuration from the Cache Unit agent**, so it tracks the DUT exactly, including stall timing.

5.1 State

`PC` (7 bits), active IMEM bank, active CR bank, `LR[16]` (32 bit), `CR[2][16]` with `CR0=0`, `CR1=1`, `halted`, `imem_rd_sel`.

5.2 Per-cycle behaviour

Step	Model action
Fetch	`inst_vliw = imem[active][PC]`. Non-branch next word = `PC+1`. On a branch, both `PC+1` and the target are candidates; `taken` selects. No bubble is ever inserted for a branch.
Snapshot reads	Branch operands and LR-lane sources are read from the values committed at the end of the previous cycle.
COND	`BEQ / BNE / BLT.S / BGE.S` evaluate; `BR` is unconditional, target = low 7 bits of the register; `HALT` stops fetch (PC and word hold).
LR lanes	Three lanes compute in parallel from the snapshot (32-bit wrap; `INC.MODP2` mask; `ADDB/ADDBI` per-byte saturate [0,255]) and commit at end of cycle.

Forwarding	The four buses carry the **post-write** resolved operands. `xmem_read_addr = CR[base] + LR[offset]` from the post-write LR; `xmem_read_en = load != NOP`.
APB	Register map, inactive-bank writes, swap effect on the next fetch cycle; `pslverr` on write to `CR0/CR1`, read-only registers, or unmapped addresses; the LR file is not writable over APB.
Stall (Q1)	The Cache Unit agent is the source of truth for when a stall starts and ends, and reports stall events to the model. During a stall the model holds PC and word, holds the buses and drives `xmem_read_en` low in the bubble. Separately, the agent self-checks every stall decision against the CRM tag rule.

For the probe scoreboard the model also emits a predicted `probe_item` every cycle (PC, next PC, word, branch result, LR writes, LR file, bank IDs). The model also emits per-word metadata (ops, branch taken, hazard flags, loop iteration) to the coverage subscriber and a look-ahead stream (upcoming load addresses) to the Cache Unit agent.

6. Scoreboards

6.1 Comparison mechanism

Each scoreboard holds an expected queue (from the reference model) and an observed queue (from the monitor), compares per cycle, and reports the first mismatch with cycle, PC and word. Timing-only changes (stalls) must not change any compared value except the cycle alignment the reference model already accounts for.

6.2 Scoreboard instances

Scoreboard	Predicted vs observed
Bus scoreboard	All four VLIW buses, each cycle: slot fields and resolved CR/LR operands (post-write values)
XMEM scoreboard	`xmem_read_en` and `xmem_read_addr` each cycle
Probe scoreboard	Internal state every cycle: `PC`, `next_pc`, `inst_vliw`, `branch_taken`, the three LR write lanes, the full LR file, active bank IDs, `fetch_stopped`

6.3 Assertions (SVA, bound to the DUT)

- (Bound on the probe interface.) After a branch word, the next cycle's word is the one at `taken ? target : PC+1` (no bubble).
- After `HALT`, no new word is fetched until the next kernel starts.
- `xmem_read_en` is high only when the load slot is non-NOP.
- `CR0 == 0` and `CR1 == 1` always.
- No APB wait cycles caused by IPU activity (`pready` depends only on the slave write port).

7. Functional Coverage

7.1 Word-level coverpoints

- Each LR opcode × each lane (0, 1, 2); 0–3 lanes active in one word.
- `SET` mode (CR vs. immediate; immediate −8, −1, 0, 7); `INC.MODP2` k ∈ 1..9; `ADDB/ADDBI` saturation at 0 and 255; 32-bit wrap in `ADD/SUB/INC/DEC`.
- Each COND op, taken and not taken; branch operands LR vs. CR; `BR` source LR vs. CR; `BR` with junk upper bits.
- Load: NOP vs. active; base CR index; offset LR index.

7.2 Scenario-level coverpoints

- Trace shape (seen at run time): loop count, nesting, one-word loop, `BR` target distance, backward vs. forward branch, loop iteration counts.
- Branch in consecutive words; branch to `PC+1`; branch to itself; branch to word 0 and word 127; program length (1, 2, 64, 127, 128 words).

- Guard exit by guard vs. by data condition; `DEC` before vs. in the branch word.
- Back-to-back kernels; preload during run; swap immediately after HALT; APB traffic during execution.
- APB: each error type; gap mode; read during run.

7.3 Required cross coverage

- **Snapshot hazards (sampled from the probe):** LR written in word N × read by {lane source, branch operand, each bus operand, load offset} in the same word.
- **Stall × event:** {each fill mode} × {load in branch word, load after LR write, last word before HALT, first word after taken branch, back-to-back loads, first access of a bank, `jump_back` boundary}.
- **Cache-state events:** tag miss then release; bank freed then re-read on a new repetition; all 16 banks used (wrap of `offset[19:10] % NUM_BANKS`); free-all on `offset == FFFF`; stall length buckets (0, short, bank-fill length).
- COND op × taken × same-word LR write to a branch operand.
- Kernel kind × fill mode × APB gap mode.

8. Key Parameters

Parameter	Default	Note
`REG_WIDTH`	32	CR/LR width (the spec's block diagram says 20 — Q3)
`IMEM_BANKS / IMEM_DEPTH / IMEM_BANK_DEPTH`	2 / 256 / 128	PC width 7 bits
`IMEM_BANK_READ_PORTS`	2	2R1W per bank: dual fetch on branch
`CR_BANKS / CR_REG_COUNT`	2 / 16	`CR0=0`, `CR1=1` hard-wired
`LR_REG_COUNT / LRD_REG_COUNT / LR_LANES`	16 / 8 / 3	LRD = register-pair aliases
`COND_OP_COUNT / LR_OP_COUNT`	6 / 8	COND: 5 branches + HALT
`APB_ADDR_W / APB_DATA_W`	32 / 32	
`NUM_BANKS / rows per bank / META_DATA_WIDTH`	16 / 1024 / 16	Cache Unit; tag = {table_id, offset[19:10]}
`XMEM_ADDR_W`	impl	Truncation of CR+LR sum: Q5
`*_BUS_W`	impl	Layout needed for scoreboard: Q6
`lock`-style test knobs	—	`test_type, program_mode, num_kernels, requested length T`

9. Open Questions and Spec Issues

#	Issue	Impact on the plan
Q1	The Cache Unit (CRM) stalls the IPU on a tag miss, but CTRL has **no stall / ready input** and its §9 says it does no interlocks (yet §7.1 mentions a "bubble"). Which signal carries the stall? Also: CRM needs `table_id` with the address; CTRL outputs only `xmem_read_addr`.	Defines the Cache Unit agent's output signal and the reference model's stall behaviour. **Highest priority.**
Q2	ADR 0007 removed the STORE slot; the Control Stage page still has `str_vliw_bus` (action A-4).	Plan keeps `HAS_STORE_BUS` as a parameter.
Q3	Block diagram says CR/LR are 20-bit; the parameters and text say 32-bit.	Plan uses 32.
Q4	How a kernel starts and how completion is signalled is not on the port list (`kernel_done` is mentioned in prose; the bank-swap is described as both an APB register and an external `imem_bank_sel`).	Start = swap register write (assumed). End = reference model HALT + monitors, until a status register is defined.
Q5	`XMEM_ADDR_W` vs. the 32-bit `CR + LR` sum: truncate or error?	Reference model address width.

Q6	Bus layouts (`*_BUS_W`), APB address map and the IMEM word width (decoded VLIW is wider than 32 bits, so multiple APB writes per word?) are not defined.	Needed for `bus_layout` and `apb_load_program_seq`.
Q7	Spec text says "11 mnemonics" and "seven branches"; the tables list 8 LR ops and 6 COND ops (5 branches + HALT). §10.1 refers to "see §10" for the same-destination rule.	Plan follows the tables.
Q9	The RTL hierarchy and names of the internal signals are not known. The spec lists them only in a note (§3) and says they do not cross the stage boundary; `CR/LR` file storage layout and the bank-select registers are also not described.	Needed for the single `ctrl_probe_if` mapping file.
Q10	Are the LRs cleared at kernel start? In a back-to-back test the next kernel may start with the previous kernel's LR values.	Starting state for the generator's stop check.
Q8	The Cache Unit itself is not under test. Its behaviour is modelled by the agent from the spec (tag rule, bank release, `jump_back`). The spec has a few inconsistencies (e.g. `FFFF` used both as a 14-bit tag and as an offset; `bank.tag = i` in §9.2) that the model must resolve.	Agent model details.

10. Milestones

- **M1** — APB agent with clock plumbing, probe agent + `ctrl_probe_if` mapping, directed program load, reference model for non-branch words, bus and probe scoreboards.
- **M2** — Execution-guided generator with the state tracker (LR ops, COND ops, undo on loops), offline check of the tracker against the reference model, branch model, no-bubble assertions.
- **M3** — Loads with legal table addresses, `BR` computed jumps, `LRD` pairs, hazard knobs, APB error injection.
- **M4** — Cache Unit agent (tag model + fill modes, after Q1 is closed), load-address planner in the generator, REAL test (back-to-back kernels).
- **M5** — Coverage closure and seed regression.